

100 Days of Azure DevOps

Day 56 Handout — A one-node cluster is still a control plane plus a VM bill — create only with a destroy date

6 - Containers & Kubernetes · Personal lab only · Educational content · Not a sales pitch

Architecture A — control plane, node pools, kube-system vs your pool

Piece	What Azure runs	What you must not confuse
Control plane	Managed API server, etcd, scheduler. You pay the AKS SKU relationship	A VM you SSH to 'the master'. You do not
System node pool	First pool, mode=System. kube-system DaemonSets/pods (CoreDNS, CoreDNS-autoscaler, CSI)	A place to schedule myapp 'because there is only one node'
User node pool	mode=User. Where Deployments for workloads should land (taints/labels in real designs)	Optional on a 1-node lab — still know the mode exists
Network plugin	kubenet (pod CIDR + NAT) vs Azure CNI (pod VNet IPs) vs Azure CNI Overlay (recommended IP conservation)	Rebuilding the cluster three times to 'compare' as a personality trait
Identity	kubelet uses managed identity (kubelet_identity) for ACR AcrPull	imagePullSecrets with ACR admin as the default

Architecture B — create, skip, or Container Apps

Constraint	Default	Create AKS only when
Cost / attention	Skip AKS. Use kind + ACI/Container Apps path	You can name this weekend's az aks delete date before create returns
Need kube API in Azure	Smallest: --node-count 1, cheap VM size, Overlay CNI	Production wants 3 system nodes. This is a lab, not that
Network plugin	Azure CNI Overlay for a modern default	You can explain kubenet IP vs CNI subnet exhaustion in one sentence
SSH keys	--generate-ssh-keys is fine	Do not copy those keys into the Git repo
Skip is engineering	Write why in docs — Container Apps / ACI	Skipping is cheaper than a forgotten node in November

Architecture C — pitfalls that keep billing after the weekend demo

Pitfall	What you see	Fix
No destroy date	'Lab cluster' still in Cost Analysis next quarter	Calendar before create. az aks delete --yes --no-wait
Apps in kube-system	CrashLoop next to CoreDNS; later NetworkPolicy stories	namespace workloads; system pool is plumbing
Default NSG folklore	Cannot pull ACR / cannot hit API	Attach AcrPull to kubelet identity; az aks get-credentials --overwrite-existing
Standard_D8 'because lab'	Bill is the lesson, not kubectl	Standard_B2s or the cheapest region SKU that AKS accepts
Leave node-count 1 as architecture advice	Resume-ready cluster that is not HA	Lab only. Notes must say system pool min=3 in real life

One-liner to remember

AKS is a gym membership for orchestration — easy to start, painful if you forget to cancel. If you cannot name the destroy date, do not create the cluster.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal subscription. Optional cluster. If you create aks-lab, destroy is scheduled the same weekend. No employer cluster kubeconfig.

- 1 Read AKS node pools and kubernetes vs Azure CNI vs Azure CNI Overlay at a survey level. Write one sentence each. Do not rebuild three clusters to feel thorough.
- 2 Decide: create smallest AKS, or skip to Container Apps/ACI and write why. If you cannot put destroy on this weekend's calendar, you skip.
- 3 If creating: az group create rg-day56; az aks create with --node-count 1, a small VM size, --network-plugin azure --network-plugin-mode overlay, --generate-ssh-keys. Then az aks nodepool list and note mode=System.
- 4 az aks get-credentials -g rg-day56 -n aks-lab --overwrite-existing. kubectl get nodes. kubectl get pods -n kube-system. Do not apply myapp into kube-system.
- 5 Grant the kubelet identity AcrPull on the lab ACR if you will deploy tomorrow (az aks show --query identityProfile.kubeletidentity.objectId).
- 6 Write docs/aks-day56.md: created or skipped, destroy date or skip reason, system vs user pool, kube-system vs workloads. If created, put az aks delete in the doc as a copy-paste.

Starter CLI — smallest cluster, Overlay CNI, delete is documented

```
# Cost warning: AKS is not a daily-delete toy. Calendar destroy first.
az aks create -g rg-day56 -n aks-lab \
  --node-count 1 --node-vm-size Standard_B2s \
  --network-plugin azure --network-plugin-mode overlay \
  --generate-ssh-keys --enable-managed-identity
az aks nodepool list -g rg-day56 --cluster-name aks-lab -o table
az aks get-credentials -g rg-day56 -n aks-lab --overwrite-existing
kubectl get nodes -o wide
kubectl get pods -n kube-system
# Destroy (same weekend):
# az aks delete -g rg-day56 -n aks-lab --yes --no-wait
```

Done checklist

- Either a one-node lab cluster with a written destroy date, or a written skip to ACI/Container Apps
- Can explain system node pool vs user pool and why myapp is not kube-system
- Can explain Overlay vs kubernetes vs CNI in one sentence each
- docs/aks-day56.md exists; no SSH keys in Git

Tomorrow — Day 57

Deploy to AKS from a pipeline (or the same manifests against kind)